



Higher order ordinary differential equations

Linear ordinary differential equations of higher order.

Solution of ordinary differential equations of higher order.

Solution of second order boundary value problems (BVPs)

The solution of a second-order BVP is a function that is twice differential, satisfies the given differential equation, and also satisfies the given boundary conditions. For example, consider the following BVP

$$\frac{d^2 y}{dx^2} + y = 0; \quad y(0) = 2, \quad y\left(\frac{\pi}{2}\right) = -3. \quad (5)$$

It could easily be observed by substituting into the above differential equation that

$$y(x) = A \cos x + B \sin x$$

is a solution of the differential equation for any real constants A and B . Using the boundary condition $y(0) = 2$, we get from equation (5) that $A = 2$. Similarly, using $y\left(\frac{\pi}{2}\right) = -3$, we get from (5) that $B = -3$.



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Finally, substituting the values of arbitrary constants A and B in (5), we obtain the solution of the given BVP as

$$y(x) = 2 \cos x - 3 \sin x.$$

Operator notation

Introducing $\frac{d}{dx} = D$ (differential operator, i.e., an operation of differentiation), we have

$$\frac{dy}{dx} = Dy, \quad \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = D^2y, \quad \dots, \quad \frac{d^ny}{dx^n} = D^ny.$$

The second-order linear differential equation of the form

$$\frac{d^2y}{dx^2} + P_1 \frac{dy}{dx} + P_2y = Q,$$

where P_1 , P_2 , and Q are functions of x alone (or constants) and y is unknown function of x , can be written as



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$$D^2y + P_1Dy + P_2y = Q,$$

$$(D^2 + P_1D + P_2)y = Q,$$

$$F(D)y = Q.$$

Where, $F(D) = D^2 + P_1D + P_2$ is called an operator polynomial in D .

Example:-

The following differential equation

$$\frac{d^2y}{dx^2} + x^2 \frac{dy}{dx} - 2xy = \sin x$$

can be written as

$$(D^2 + x^2D - 2x)y = \sin x.$$



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Linear operator

An operator L is called a linear operator if for any constants A and B and any functions $y_1(x)$ and $y_2(x)$, we have

$$L(Ay_1 + By_2) = AL(y_1) + BL(y_2).$$

Principle of superposition

Theorem:-

Prove that if $y_1(x)$ and $y_2(x)$ are two linearly independent solutions of the second order homogeneous linear differential equation $F(D)y = 0$, then

$$y = Ay_1(x) + By_2(x),$$

where A and B are arbitrary constants, is the general solution of $F(D)y = 0$.



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Example:- Solve the following differential equation

$$\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 3y = 0.$$

Solution:-

The given differential equation can be written as

$$D^2 y + 4Dy + 3y = 0, \quad \because D = \frac{d}{dx}$$

$$(D^2 + 4D + 3)y = 0,$$

$$F(D)y = 0,$$

where, $F(D) = D^2 + 4D + 3$. The characteristics (or auxiliary) equation of the given differential equation is

$$F(m) = m^2 + 4m + 3 = 0$$

$$m^2 + 3m + m + 3 = 0$$



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$$\begin{aligned}m(m+3) + 1(m+3) &= 0 \\(m+3)(m+1) &= 0 \\m+3 = 0, m+1 &= 0 \\m &= -1, -3.\end{aligned}$$

Since the roots are real and distinct, so the general solution of the given differential equation is

$$y = c_1 e^{-1x} + c_2 e^{-3x}.$$

Example:- Solve the following differential equation

$$\frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + 6y = 0.$$

Exercise (Ref. Book No. 1, Problem set 2.2, P. No. 59, Question No. 1-15).



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Solution of non-homogeneous linear differential equations of higher order

Recall that, a linear differential equation of order n in the dependent variable y and independent variable x is of the form

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = F(x), \quad (6)$$

where, $a_n, a_{n-1}, a_{n-2}, \dots, a_2, a_1, a_0$, and $F(x)$ are functions of the independent variable x only and a_n is not identically zero. If $F(x) \neq 0$, then the equation (6) is called as **non-homogeneous** or **complete** equation. If $F(x) = 0$ (i.e., $F(x) = 0$ for all x considered), then equation (6) becomes

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = 0 \quad (7)$$



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and is called **homogeneous**, **reduced**, or **complementary** equation.

For example

$$\frac{d^2 y}{dx^2} + x^2 \frac{dy}{dx} - 2xy = \sin x$$

is a second-order non-homogeneous linear differential equation, whereas

$$\frac{d^2 y}{dx^2} + x^2 \frac{dy}{dx} - 2xy = 0$$

is called the corresponding second-order homogeneous linear differential equation.

Recall also that, a solution of a differential equation which contains as many arbitrary constants as the order of the differential equation is called as general solution (or integral) and the solution obtained from the general solution by giving particular values to arbitrary constants is called as particular solution.



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Remarks:- Before we investigate the solution of (6), we have to keep an eye on the following vital aspects:

- 1 Every n -th order homogeneous linear differential equation of the form

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = 0$$

has n linearly independent solutions, i.e., $y_1, y_2, y_3, \dots, y_n$.

- 2 If $y_1, y_2, y_3, \dots, y_n$ are n linearly independent solutions of equation (7), then any linear combination

$$y_c = c_1 y_1 + c_2 y_2 + c_3 y_3 + \cdots + c_n y_n$$

is the general solution of equation (7). Where, $c_1, c_2, c_3, \dots, c_n$ are arbitrary constants.

- 3 Let y_p be any particular solution of equation (6), i.e., y_p does not contain any arbitrary constant. Then $y_c + y_p$ is called as the general solution of equation (6).



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Thus in order to find the general solution of equation (6), we have to obtain the set of n independent solutions of (7) to determine y_c and a particular solution y_p of equation (6) and then

$$y = y_c + y_p$$

is the general solution of equation (6). Where y_c is called the **Complementary Function (C. F.)** and y_p is called the **Particular Integral (P. I.)**.



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The method of undetermined coefficients

In order to find particular integral y_p using the method of undetermined coefficients, we have to remember the following table where we list expression(s) of y_p when $F(x)$ is function of a particular type:

$F(x)$	y_p
a	$x^k A$
ax^n	$x^k (A_0 x^n + A_1 x^{n-1} + A_2 x^{n-2} + \dots + A_{n-1} x + A_n)$
ae^{rx}	$x^k (Ae^{rx})$
$a \cos(ax)$ or $a \sin(ax)$	$x^k (A \cos(ax) + B \sin(ax))$

where k is the smallest non-negative integer which will ensure that no-term in the particular integral y_p is already in the complementary function y_c .



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Note:-

If $F(x)$ is the sum of several terms, then we have to write y_p for each term individually and then add-up all of them.

Example:-

Solve the following differential equation

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = 2x^3 - 9x^2 + 6x.$$

Solution:-

The given differential equation can be written as

$$D^2 y - 3Dy + 2y = 2x^3 - 9x^2 + 6x, \quad \because D = \frac{d}{dx}$$

$$(D^2 - 3D + 2)y = 2x^3 - 9x^2 + 6x,$$

$$F(D)y = 2x^3 - 9x^2 + 6x.$$



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For the complementary function y_c , we take

$$F(D)y = 0,$$

where, $F(D) = D^2 - 3D + 2$. The characteristics (or auxiliary) equation of the above homogeneous differential equation is

$$F(m) = m^2 - 3m + 2 = 0$$

$$m^2 - 2m - m + 2 = 0$$

$$m(m - 2) - 1(m - 2) = 0$$

$$(m - 2)(m - 1) = 0$$

$$m - 2 = 0, m - 1 = 0$$

$$m = 1, 2.$$

Since the roots are real and distinct, so the complementary function of the given non-homogeneous linear differential equation is

$$y_c = c_1 e^x + c_2 e^{2x}.$$



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Now, in-order to find the particular integral y_p of the given differential equation, consider

$$y_p = x^k(Ax^3 + Bx^2 + Cx + D).$$

Since no-term of y_p is already in y_c , so put $k = 0$ and we have

$$y_p = Ax^3 + Bx^2 + Cx + D$$

Take

$$y_p' = 3Ax^2 + 2Bx + C$$

$$y_p'' = 6Ax + 2B.$$

Substituting these value in the given differential equation, we have

$$(6Ax + 2B) - 3(3Ax^2 + 2Bx + C) + 2(Ax^3 + Bx^2 + Cx + D) = 2x^3 - 9x^2 +$$



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Simplifying and then comparing coefficients, we get

$$A = 1, \quad B = 0, \quad C = 0, \quad D = 0.$$

Substitute these value in

$$y_p = Ax^3 + Bx^2 + Cx + D,$$

we have

$$y_p = x^3.$$

Hence the general solution of the given differential equation is

$$\begin{aligned} y &= y_c + y_p \\ y &= c_1 e^x + c_2 e^{2x} + x^3. \end{aligned}$$

Exercise (Ref. Book No. 1, Problem set 2.7, Question No. 1-18).



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Method of Variation of Parameters

The method of variation of parameters due to French mathematician J. L. Lagrange (1736 - 1813) is a general method for finding solution of n -th order non-homogeneous linear differential equation of the form

$$\frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + P_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + P_{n-2} \frac{d^2 y}{dx^2} + P_{n-1} \frac{dy}{dx} + P_n y = Q.$$

For simplicity, we consider the following second-order non-homogeneous linear ODE

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = F(x), \quad (8)$$

where P , Q , and F may be variable (given functions of x), but we assume that they are continuous on some open interval I .

Previously, we have seen that a general solution of equation (8) is the sum of a general solution y_c , i.e., complementary function of the corresponding homogeneous ODE and any particular solution y_p of (8).



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To obtain y_p when $F(x)$ is not too complicated, we often use the method of undetermined coefficients. However, since this method of undetermined coefficients is restricted to functions $F(x)$ whose derivatives are of a form similar to $F(x)$ itself (powers, exponential functions, etc.), it is desirable to have a method valid for more general ODE (8) called the method of variation of parameters. To use this method, it is essential to know the general solution of the corresponding homogeneous equation

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0. \quad (9)$$

Suppose that $y = y_1(x)$ and $y = y_2(x)$ are two linearly independent solutions of equation (9). Then the complementary function y_c of equation (8) is given by

$$y_c = c_1 y_1 + c_2 y_2,$$

where c_1 and c_2 are arbitrary constants. We replace these arbitrary



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constants c_1 and c_2 by unknown functions $u_1(x)$ and $u_2(x)$, respectively, and require that

$$y_p = u_1 y_1 + u_2 y_2, \quad (10)$$

be a particular solution of equation (8). Basically, the method of variation of parameters involves in replacing two arbitrary constants c_1 and c_2 by the two unknown functions $u_1(x)$ and $u_2(x)$ to be determined such that the resulting function in (10) becomes a particular integral of equation (8). This replacement of constants or parameters by variables gives the method its name, i.e., the method of variation of parameters.

Now differentiating equation (10) w. r. t. x , we have

$$y_p' = (u_1' y_1 + u_2' y_2) + u_1 y_1' + u_2 y_2'. \quad (11)$$

Since, equation (10) contains two unknown functions u_1 and u_2 , so in order to determine them we need two conditions. However, one of these conditions is that equations (10) must satisfy equation (8). Thus



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we are at the liberty to impose arbitrarily the second condition. We choose this condition to be the one which satisfies equation (11) most, namely

$$u_1' y_1 + u_2' y_2 = 0. \quad (12)$$

Basically, if we differentiate equation (11) then y_p'' will contain u_1'' and u_2'' , so in order to avoid second-derivatives of u_1 and u_2 we impose the above condition.

Then equation (11) reduces to

$$y_p' = u_1 y_1' + u_2 y_2'. \quad (13)$$

Differentiating equation (13), we have

$$y_p'' = u_1' y_1' + u_2' y_2' + u_1 y_1'' + u_2 y_2''. \quad (14)$$



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Substituting the values of y_p , y_p' , and y_p'' into equation (8) and collecting terms containing u_1 and terms containing u_2 , we have

$$u_1(y_1'' + Py_1' + Qy_1) + u_2(y_2'' + Py_2' + Qy_2) + u_1'y_1 + u_2'y_2 = F(x).$$

The expressions within parentheses are zero because, y_1 and y_2 are solutions of equation (9). Thus the above equation reduces to

$$u_1'y_1 + u_2'y_2 = F(x). \quad (15)$$

This gives a second equation in u_1' and u_2' . Now the equations (12) and (15), i.e.,

$$\begin{aligned} u_1'y_1 + u_2'y_2 &= 0, \\ u_1'y_1 + u_2'y_2 &= F(x), \end{aligned}$$

form a system of two linear algebraic equations in two unknowns u_1' and u_2' . The solution of this system of equations can be obtained by



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using Cramer's rule as

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ F(x) & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}} = -\frac{y_2 F(x)}{W}, \quad u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & F(x) \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}} = \frac{y_1 F(x)}{W},$$

where,

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2 \neq 0$$

is the Wronskian of y_1 and y_2 . The unknown functions $u_1(x)$ and $u_2(x)$ can then be obtained by integrating u_1' and u_2' , respectively, i.e.,

$$\left. \begin{aligned} u_1(x) &= - \int \frac{y_2 F(x)}{y_1 y_2' - y_1' y_2} dx = - \int \frac{y_2 F(x)}{W} dx, \\ u_2(x) &= \int \frac{y_1 F(x)}{y_1 y_2' - y_1' y_2} dx = \int \frac{y_1 F(x)}{W} dx. \end{aligned} \right\} \quad (16)$$



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We obtain the particular integral by substituting $u_1(x)$ and $u_2(x)$ in equation (10) and finally general solution of equation (8) is given by

$$y = y_c + y_p.$$

Note:- If the constants of integration in equations (16) are left arbitrary, then equation (10) represents the general solution of equation (8).

Summary of the method:- In numerical problems, instead of performing the complete process, we can work directly to evaluate $u_1(x)$ and $u_2(x)$. To find the general solution of equation (8), the steps are as follows:

- 1 Find the complementary function y_c and write the functions y_1 and y_2 .
- 2 Calculate W and then $u_1(x)$ and $u_2(x)$ from equation (16).
- 3 Substitute $u_1(x)$ and $u_2(x)$ in equation (10) to find the particular integral and then write the general solution of equation (8) as $y = y_c + y_p$.



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Example:- Solve the following non-homogeneous linear ODE by variation of parameters. Show the detail of your work

$$\frac{d^2 y}{dx^2} + 9y = \sec 3x.$$

Solution:- The given differential equation can be written as

$$\begin{aligned} D^2 y + 9y &= \sec 3x, & \therefore D &= \frac{d}{dx} \\ (D^2 + 9)y &= \sec 3x, \\ F(D)y &= \sec 3x. \end{aligned}$$

For the complementary function y_c , we take

$$F(D)y = 0,$$

where, $F(D) = D^2 + 9$. The characteristics (or auxiliary) equation of the above homogeneous differential equation is



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$$F(m) = m^2 + 9 = 0$$
$$m = 0 \pm 3i.$$

Since the roots are complex, so the complementary function of the given non-homogeneous linear differential equation is

$$y_c = e^{0x}(c_1 \cos 3x + c_2 \sin 3x),$$

$$y_c = c_1 \cos 3x + c_2 \sin 3x,$$

$$y_c = c_1 y_1 + c_2 y_2,$$

where $y_1 = \cos 3x$ and $y_2 = \sin 3x$. Now, in-order to find particular integral y_p by variation of parameters, consider

$$y_p = u_1 y_1 + u_2 y_2.$$



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Take

$$\begin{aligned}W &= y_1 y_2' - y_1' y_2 \\&= \cos 3x(3 \cos 3x) - \sin 3x(-3 \sin 3x) \\&= 3 \cos^2(3x) + 3 \sin^2(3x) \\&= 3 \neq 0.\end{aligned}$$

Since W is non-zero, so we can calculate u_1 and u_2 as

$$\begin{aligned}u_1(x) &= - \int \frac{y_2 F(x)}{W} dx, \\&= - \int \frac{\sin 3x \sec 3x}{3} dx, \quad \because F(x) = \sec 3x \\&= - \frac{1}{3} \int \frac{\sin 3x}{\cos 3x} dx, \\&= \frac{1}{9} \int \frac{-3 \sin 3x}{\cos 3x} dx = \frac{1}{9} \ln(\cos 3x).\end{aligned}$$



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$$\begin{aligned}u_2(x) &= \int \frac{y_1 F(x)}{W} dx, \\&= \int \frac{\cos 3x \sec 3x}{3} dx = \frac{1}{3} \int dx = \frac{1}{3}x.\end{aligned}$$

Thus the particular integral is

$$\begin{aligned}y_p &= u_1 y_1 + u_2 y_2, \\&= \left(\frac{1}{9} \ln(\cos 3x)\right) \cos 3x + \left(\frac{1}{3}x\right) \sin 3x.\end{aligned}$$

Hence the general solution of the given differential equation is

$$y = y_c + y_p$$

$$y = c_1 \cos 3x + c_2 \sin 3x + \left(\frac{1}{9} \ln(\cos 3x)\right) \cos 3x + \left(\frac{1}{3}x\right) \sin 3x.$$

Exercise (Ref. Book No. 1, Problem set 2.10, Question No. 1-13).